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Growth variability of mackerel (*Scomber scombrus*) off north and northwest Spain and a comparative review of the growth patterns in the northeast Atlantic

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Abstract

This paper studies the age, male and female growth patterns and the interannual growth variability during the 1990–2000 period of the northeast Atlantic mackerel (*Scomber scombrus* Linnaeus, 1758) off north and northwest Spain (ICES Divisions VIIIc and IXa north). Age was determined by interpreting and counting growth rings on the otoliths (*sagitta*) of 7732 individuals caught during the spawning seasons by the commercial fleet and during fisheries research surveys. The parameters of the von Bertalanffy growth curve were estimated for sex combined for the whole period studied ($L_\infty = 42.7$ cm, $k = 0.268$ per year, $t_0 = -2.17$ years) and for each year; for males ($L_\infty = 42.1$ cm, $k = 0.276$ per year, $t_0 = -2.15$ year) and females ($L_\infty = 43.0$ cm, $k = 0.264$ per year, $t_0 = -2.21$ year) for the whole period studied. The values of L_∞ and k of the growth curves in males and females exhibited significant differences (likelihood ratio test, $P < 0.05$) but not t_0 ($P = 0.214$). Length distributions by age group were compared by applying non-parametric tests (Kruskal–Wallis and Kolmogorov–Smirnov). Significant differences were found ($P < 0.05$) between the years analysed, with high variability up to age 7. There were significant differences between sexes in some length distributions at age (Kolmogorov–Smirnov, $P < 0.05$, ages 4–7, 9 and 11). The information obtained in this study was compared statistically with the available growth data for the mackerel in other geographic areas in the northeast Atlantic. The growth pattern showed variability along the geographical distribution range of the species. Spatio-temporal variability in the growth of mackerel would be taking into account in the stock assessment. The statistical differences between sexes found in growth pattern seem not implicate different behaviour, and a sex-specific assessment is not justified.

Keywords: Age; Growth; Mackerel; Northeast Atlantic; *Scomber scombrus*

1. Introduction

The northeast Atlantic mackerel (*Scomber scombrus*) is an abundant pelagic species distributed from the south of the Iberian Peninsula (ICES Division IXa) to the north of Norway (Division IIa) (Fig. 1).

For assessment and management purposes mackerel is currently considered to be a single stock divided into three spawning components (ICES, 1996, 2000, 2004): the western component, made up of individuals that spawn in western European waters (ICES Subareas VI, VII and Divisions VIIIabde), the south-

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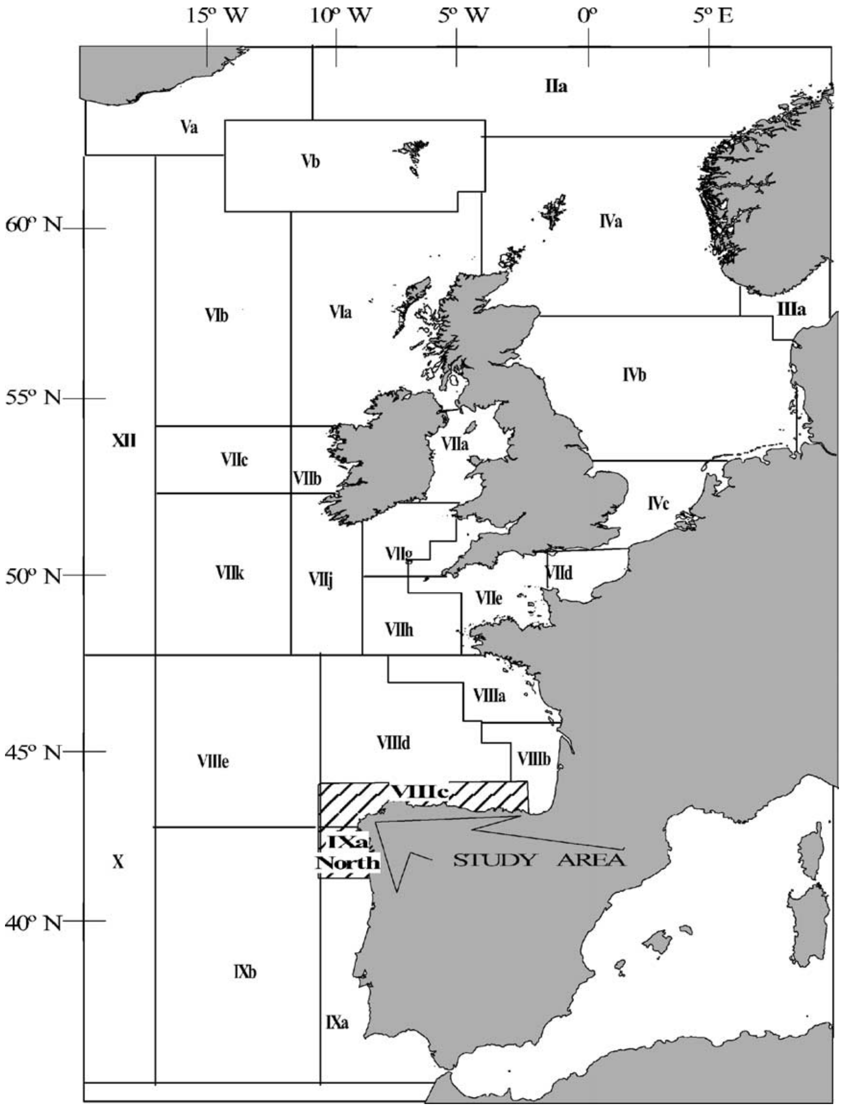


Fig. 1. Location of the study area and ICES Subareas and Divisions in the northeast Atlantic.

ern component, which includes individuals spawning in southern European waters (VIIIc and IXa) and the North Sea component, whose spawning areas are in the North Sea and Skagerrak (IIIa and IV). Mackerel is a migratory species. After spawning, adults belonging to the southern and western components migrate to the feeding areas, the Norwegian Sea and the North Sea, during the second half of each year, returning to the spawning areas at the beginning of the following year (Rankine and Walsh, 1982; Iversen and Skagen, 1989; Uriarte and Lucio, 2001; Uriarte et al., 2001). This species supports valuable fisheries of great importance in several European countries, with annual mean landings of 675,000 t from 1972 to 2002 (ICES, 2004).

Several fleets and gears seasonally exploit the mackerel resource in southern European waters: handline, purse seine, trawl and gillnet. Catches have increased in recent years, and in 1998 and 1999 they reached a peak of 44,000 t, of which 38,000 t were in Division VIIIc (ICES, 2001). Adult mackerel are mainly taken as target species by handline in Division VIIIc. Juveniles are caught in the second half of the year, mostly in Division IXa (Villamor et al., 1997).

Ageing and growth parameter estimation are fundamental prerequisites for stock assessment. A great number of growth studies on mackerel have been carried out, given the importance of their fisheries. The growth of North Sea and Skagerrak mackerel was studied by Kandler (1957), Nedelec (1960), Aker (1961), Castelló and Hamre (1969), Postuma (1972) and Skagen (1989); growth in the western mackerel, by Steven (1952), Molloy (1964), Bolster (1974), Lockwood and Johnson (1976), Kästner (1977), Corten and Van de Kamp (1978), Eltink and Gerritsen (1982) and Skagen (1989), in the south by Gordo et al. (1982), Gordo and Martins (1984), Cort et al. (1986), De la Hoz and Villegas (1987), Lucio (1997) and Martins (1998). Nevertheless, none of the above carried out a comparative analysis of growth taking the year factor into account nor did they address statistical differences in the growth pattern between sexes.

There have been descriptions of different growth groups in the northeast Atlantic mackerel, which in some cases point to the existence of different subpopulations to the west of the British Isles (Kästner, 1977; Corten and Van de Kamp, 1978) and others that have put forth the alternative hypothesis that the growth differences in the mackerel may be attributed to the changes in length at age observed after they have migrated (Eltink and Gerritsen, 1982; Dawson, 1986; Eltink, 1987).

This paper examines the growth pattern of southern component mackerel and its interannual variability, taking into account a wider geographical area and a greater number of years than the studies previously cited for this component. We have analysed the differences in the growth pattern between males and females and present a statistical comparison of the growth parameters found in this work with those presented by other authors reporting on other areas of the northeast Atlantic. Lastly there is a discussion on the causes of temporal and geographic variability in the growth of the mackerel.

2. Material and methods

2.1. Sampling and ageing

During the period from 1990 to 2000 biological analysis was carried out on 7732 mackerel from the north and northwest of Spain (Divisions VIIIc and IXa north). Aiming to get a complete representation of all the size groups caught in the area, monthly samples (by 100 fish each) were selected at random from the handline and purse seine commercial catches. In addition, samples were taken using pelagic mid-water trawl during the annual acoustic survey in that area. The data analysed in this study refer to samplings that were performed in the first half of each year (primarily March and April), owing, on the one hand to the marked seasonality of the fishery—as it was impossible to obtain enough representative samplings throughout the whole year—and, on the other hand, to avoid seasonal differences in the samples. Most mackerel samples were analysed fresh. However, if this was not possible, they were frozen. In the latter case, due to the shrinkage of the tissues during the freezing process, correction factors for length (L) and weight (W) were applied [P. Lucio, AZTI Sukarrieta, pers. comm.: $L(\text{estimated}) = 1.0085336L(\text{thaw})$; $W(\text{estimated}) = 1.0400555W(\text{thaw})$ for fish <25 cm, and $W(\text{estimated}) = 1.0098042W(\text{thaw})$ for fish >25 cm].

Fish were measured (total length to the lower cm), weighed (wet weight, g); their sex determined by macroscopic examination of the gonads and *sagitta* otoliths removed.

Age was estimated by interpreting and counting growth rings on the otoliths. Methodological ageing procedures described in ICES (1995) were followed. Whole otoliths were mounted on a blackened background, covered with resin (Eukitt), and illuminated by reflected light. A binocular microscope was used to observe the banding pattern (20× or 40× magnifications, depending on otolith size). A specific experienced mackerel reader aged the otoliths and each otolith was read twice, on two separate occasions. The readings for a given otolith were accepted only if both agreed. Agreement between readings was high (80%). When there was a discrepancy between the two readings, a third reading was carried out. Unreadable otoliths were rejected (1.3%). The criteria used to determine age were (ICES, 1995): (1) the date of birth was considered to be 1 January; (2) each year a translucent and an opaque growth band are deposited on the otolith, and (3) the age of the fish corresponds to the number of complete translucent bands (n). The otolith edge band from fish caught in the first half of the year was counted in adults when it was translucent (n) or opaque ($n + 1$) and in juveniles when it was translucent (n). When juveniles appear on the new opaque edge (about May) the number of translucent rings (n) was counted.

The translucent rings were counted, preferably in the anterior part (rostrum) and posterior part (post-rostrum) of the otolith. When different ages were recorded in the two areas of the otolith, the older age was considered (ICES, 1995). Ages up to 18 years were recorded.

The validation of the ageing criteria for northeast Atlantic mackerel was provided in ICES (1995) by using otoliths from fish of which the actual age was known, because otolith and dates of capture were available from recaptured tagged fish. The ageing method was validated in this workshop up to age 8 and for the oldest ages (above age 8), the modal age of the ages assigned was assumed to be the actual age, as was recommended in ICES (1994).

An age-length key was elaborated by sex for the study period (1990–2000) and by sexes combined for each year. Mean lengths at age and the corresponding standard deviations were calculated. Ages up to 14 years were considered, and those that were greater were accumulated into one group (15+).

2.2. Data analysis

To estimate the growth rate of mackerel, the whole length distributions by age class were used. The total length distribution by age class were fitted to the von Bertalanffy equation, $L_t = L_\infty[1 - \exp(-k(t - t_0))]$, for the estimate of the growth curves. In this equation, L_t is the mean fish length at age t; L_∞, k and t_0 the parameters that determine the shape of the growth curve. L_∞ is defined as the asymptotic mean length; k the rate at which the curve approaches the asymptote and t_0 the age at which mean length is zero (Francis, 1995). von Bertalanffy growth curve parameters were determined by Marquardt non-linear iterative least-square regression analysis using SPSS 10.1. Growth parameters were calculated for sexes combined and each year, and for sex and set period (1990–2000).

Growth curves were compared between sexes by the likelihood ratio test following Kimura (1980).

Total length distributions at age were compared between years by means of the non-parametric Kruskal–Wallis and Kolmogorov–Smirnov tests and the between sex comparison over the entire period was done with the Kolmogorov–Smirnov test, since the conditions for the ANOVA were not fulfilled. To carry out the comparison between years, the Kruskal–Wallis test was first applied to the whole length distributions by age groups for all years. The Kruskal–Wallis test assesses the hypothesis that the different length distributions at age by year were drawn from the same distribution or from distributions with the same median and is based on ranking the variates (lengths) (Sokal and Rohlf, 1995). When significant differences were found in certain age groups, the Kolmogorov–Smirnov test was applied by pairs of years to find the years in which the differences were detected. The Kolmogorov–Smirnov test assesses the hypothesis that each pair of years was drawn from the same distribution at age. This test is sensitive to differences in the general pattern of the distributions in the two samples and is based on the unsigned differences between the relative cumulative frequency distributions of the two samples (Sokal and Rohlf, 1995). Owing to the great number of comparisons made using the Kolmogorov–Smirnov test, the Dunn–Sidak method was applied to determine the appropriate α or the experiment-wise error rate to avoid making a type-I error (Sokal and Rohlf, 1995).

The growth curve estimated in this paper (sexes combined, period 1990–2000) was also compared with other mackerel growth curves in the northeast Atlantic from other authors by means of the likelihood ratio test, following Cerrato (1990). This test requires raw growth data, which were not available in the references. In most cases, however, the mean sizes were available. To conduct the test on similar kinds of data and equivalent age ranges (Haddon, 2001), mean lengths at age of several sets of age ranges were used to fit the von Bertalanffy equations. The corresponding new estimated parameters were tested with those from other authors, which were recalculated from the referenced data (mean lengths at age) by using SPSS 10.1.

Table 1
Annual and total mean length at age (TL, cm) and standard deviation (S.D.) determined from mackerel otolith readings (*n*: number of samples, sexes combined) from ICES Divisions VIIIc and IXa

Age	1990			1991			1992			1993			1994			1995		
	TL	S.D.	*n*	TL	S.D.	*n*	TL	S.D.	*n*	TL	S.D.	*n*	TL	S.D.	*n*	TL	S.D.	*n*
1	25.2	1.50	73	25.5	1.41	7				25.1	1.18	43	25.4	0.98	44	26.1	1.40	31
2	29.2	1.88	46	30.3	0.83	13	31.9	0.98	7	30.3	2.99	9	28.8	1.57	49	30.8	1.79	44
3	33.1	1.89	47	32.2	1.11	46	33.3	1.77	94	34.1	1.29	11	32.2	2.03	55	33.6	1.40	76
4	34.9	1.59	62	34.6	1.68	55	34.5	1.57	83	35.6	1.44	113	36.3	2.19	75	35.6	1.67	49
5	36.6	1.94	86	35.4	1.28	42	36.2	1.59	87	36.5	1.68	61	37.3	1.87	148	36.8	1.74	28
6	38.0	1.76	99	37.4	1.94	24	37.5	1.59	40	37.6	1.65	81	38.5	1.69	110	38.0	2.32	40
7	40.2	1.42	23	38.6	1.36	8	38.3	1.54	22	39.2	1.80	46	39.4	1.74	83	39.3	1.76	36
8	40.5	1.93	57	38.8	1.49	8	39.2	1.99	22	39.8	2.05	22	40.1	1.46	39	40.1	1.83	43
9	41.2	1.65	55	39.6	1.96	9	39.7	1.72	9	41.4	1.32	23	40.7	1.48	30	40.5	2.22	16
10	41.0	1.59	17	41.5	1.60	8	40.1	1.14	5	40.4	2.08	18	40.9	1.86	16	41.9	1.58	17
11	41.9	2.07	5	42.5	2.00	3	40.4	1.83	9	41.3	1.19	12	42.3	2.12	12	41.4	2.25	18
12	42.2	1.89	7	44.5	1.41	2	42.1	1.82	5	43.5	1.51	8	42.9	1.14	5	43.1	2.01	10
13	46.5	1.41	2			0	40.0	0.71	2	43.3	1.09	13	41.8	1.53	3	44.1	1.52	5
14	42.5		1	43.5	1.41	2			0	43.4	1.64	8	43.0	1.76	6	44.0	1.64	6
15+	45.5		1	43.5		1	45.8	2.52	3	43.8	1.97	12	44.5		1	43.9	1.99	7
Total			581			228			388			480			676			426

Age	1996			1997			1998			1999			2000			Total 1990–2000		
	TL	S.D.	*n*	TL	S.D.	*n*	TL	S.D.	*n*	TL	S.D.	*n*	TL	S.D.	*n*	TL	S.D.	*n*
1	24.5	1.31	57	24.3	1.36	300	25.4	1.18	170	23.4	1.03	75	24.4	1.53	290	24.6	1.47	1090
2	32.2	1.37	6	28.7	1.57	160	29.8	1.00	141	30.0	1.10	53	29.0	2.22	75	29.5	1.74	603
3	33.2	1.26	58	32.9	2.34	46	32.3	1.57	167	32.2	1.41	46	32.4	1.63	289	32.7	1.71	935
4	35.2	1.56	129	35.0	1.62	254	34.7	2.05	82	34.7	1.81	432	35.2	1.92	127	35.1	1.78	1161
5	37.2	1.86	39	36.9	1.61	168	36.8	1.64	155	37.3	1.98	68	36.4	2.12	232	36.7	1.86	1114
6	39.2	2.06	22	38.1	1.78	55	37.7	1.55	128	38.1	1.63	101	38.5	1.63	103	38.0	1.75	803
7	39.7	1.84	61	39.7	1.54	69	38.5	1.73	62	38.7	1.51	65	39.1	1.36	141	39.2	1.66	616
8	40.5	1.87	35	40.1	1.63	71	40.3	1.42	36	39.7	1.99	25	39.7	1.38	66	40.0	1.74	424
9	41.1	1.83	41	41.2	1.40	75	40.2	1.73	41	41.1	2.01	19	39.5	2.03	18	40.8	1.74	336
10	41.9	2.52	20	42.0	1.50	55	41.3	1.75	38	41.8	1.29	26	41.8	2.11	17	41.5	1.80	237
11	41.9	2.06	17	42.8	2.02	33	41.6	1.68	17	42.3	1.35	18	41.9	1.50	11	42.0	1.91	155
12	42.9	2.55	10	43.5	1.64	33	41.9	1.70	20	41.8	1.36	12	42.3	1.23	10	42.7	1.80	122
13	45.5	0.00	2	43.6	1.55	8	43.2	1.15	3	43.6	1.07	7	41.7	1.30	5	43.3	1.69	50
14	43.0	0.71	2	44.3	1.10	5	43.5	2.00	3	41.5		1	42.3	0.84	5	43.3	1.48	39
15+	44.5	1.41	5	44.7	1.55	10	43.0	2.12	2	43.2	1.15	3	44.5	2.83	2	44.2	1.78	47
Total			504			1342			1065			651			1391			7732

3. Results

The northeast Atlantic mackerel otolith *sagitta* have a deposition pattern of wide opaque bands alternating with thin translucent bands, with the first opaque bands being notably wider. In total, 7732 whole otoliths were examined (3596 males, 3850 females and 286 undetermined). Fish ranged in size from 20 to 48 cm, and the oldest fish aged were estimated to be 18 years old.

The mean length at age of mackerel during the 1990–2000 period indicated that growth was rapid during the first year (24.6 cm at age 1) and promptly slowed down after this (Table 1). There were slight differences in the mean length and mean weight by age group between years, although no trends were observed throughout the period studied (Table 1, Fig. 2). The mean length by age group from the data pooled for the 1990–2000 period differed slightly between the sexes (Table 2). Females were larger than males (except in the 15+ group) with differences ranging from 0.1 to 1.2 cm.

The von Bertalanffy equations adjusted well to the data observed ($R^2 \geq 0.84$ in all cases, except in 1992, $R^2 = 0.67$, Table 3, Fig. 3). The growth parameters of the von Bertalanffy model estimated for the whole period were $L_\infty = 42.7$ cm, $k = 0.268$ per year and $t_0 = -2.17$ year for sexes combined; $L_\infty = 42.1$ cm, $k = 0.276$ per year and $t_0 = -2.15$ year for males and $L_\infty = 43.0$ cm, $k = 0.264$ per year and $t_0 = -2.21$ year for females. The likelihood ratio test indicated that male and female mackerel have different von Bertalanffy growth curves (ages 1–18: $\chi^2 = 215.2$, d.f. $= 3$, $P < 0.001$). A more detailed analysis of the test showed significant differences in L_∞ ($P < 0.001$) and k ($P = 0.005$), but not in t_0 ($P = 0.214$) (Table 4). There were significant differ-

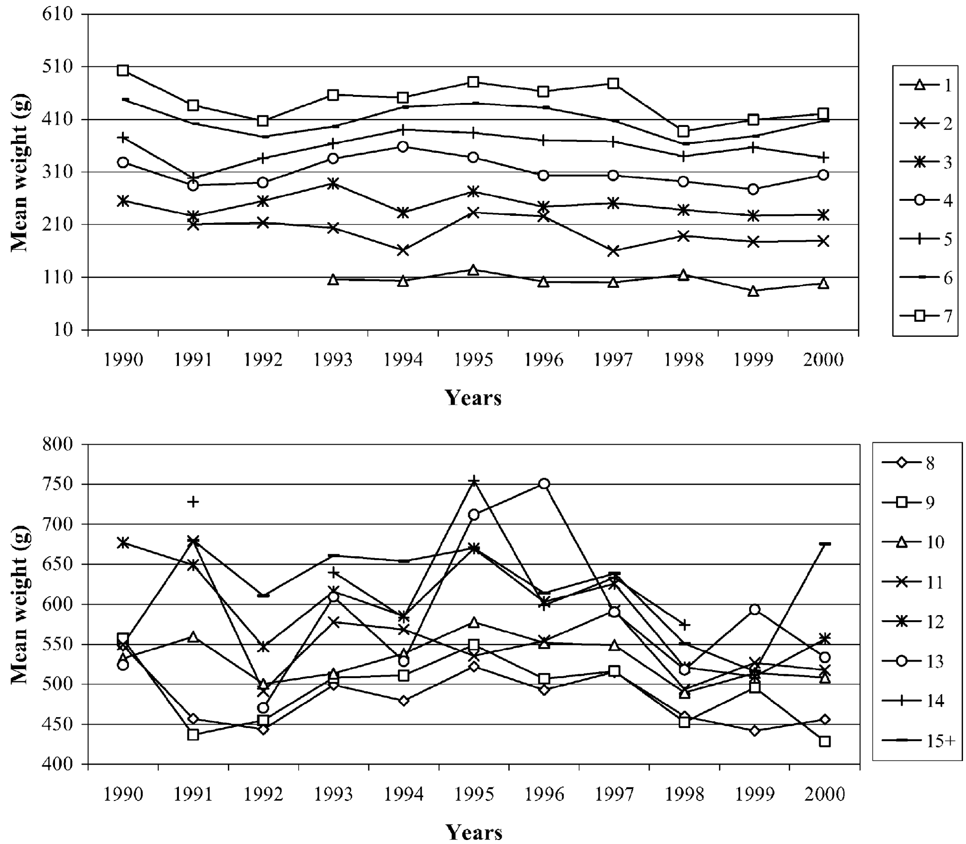


Fig. 2. Mean weight at age (g) of mackerel from ICES Divisions VIIIc and IXa, 1990–2000. Upper figure, ages 1–7 and lower figure, ages 8–15+.

Table 2
Mean length at age (TL, cm) and standard deviation (S.D.) for male and female mackerel determined from otolith readings (n: number of samples) from ICES Divisions VIIIc and IXa, sampled during 1990–2000

Age	Male			Female		
	TL	S.D.	n	TL	S.D.	n
1	24.6	1.47	487	24.7	1.48	491
2	29.4	1.79	255	29.7	1.64	298
3	32.6	1.68	462	32.7	1.71	430
4*	35.0	1.74	619	35.3	1.81	554
5*	36.5	1.87	549	37.0	1.84	554
6*	37.8	1.73	354	38.3	1.75	436
7*	39.0	1.72	275	39.3	1.60	336
8	39.8	1.79	197	40.2	1.76	223
9*	40.5	1.66	141	41.2	1.72	77
10	41.3	1.95	93	41.7	1.70	138
11*	41.4	1.94	72	42.5	1.72	77
12	42.4	1.54	41	43.0	1.83	76
13	42.4	1.30	21	43.6	1.69	31
14	43.3	1.47	17	43.4	1.55	20
15+	44.3	2.03	13	44.2	1.63	28
Total	34.7	5.34	3596	35.4	5.61	3850

* Indicates significant difference.

ences between sexes in some length distributions at age (Kolmogorov–Smirnov, $P < 0.05$, ages 4–7, 9 and 11, Table 5).

A comparison of the size distributions by age group between the years during the period under study revealed significant differences until age 10, with the greatest differences being found in the early ages. Length distributions by age group showed interannual variability (Kruskal–Wallis, $P < 0.001$ in age groups 1–7; $P < 0.01$ in age groups 8–10. Kolmogorov–Smirnov tests revealed differences between pairs of years, in 22% ($P < 0.05$) of the cases up to age 7, and in 1% ($P < 0.05$) of the cases in ages 8–10, Table 5).

The original growth parameters of the von Bertalanffy equation estimated by different authors in areas of the northeast Atlantic differ from the recalculated values, with the exception of Lucio (1997) (Tables 6 and 7). The likelihood ratio test exhibited significant differences between the growth curves estimated in our paper and those reported by other authors. When compared to Martins (1998), the test showed significant differences in the three growth parameters. As regards the other authors, the differences were significant in one or two parameters (Table 7).

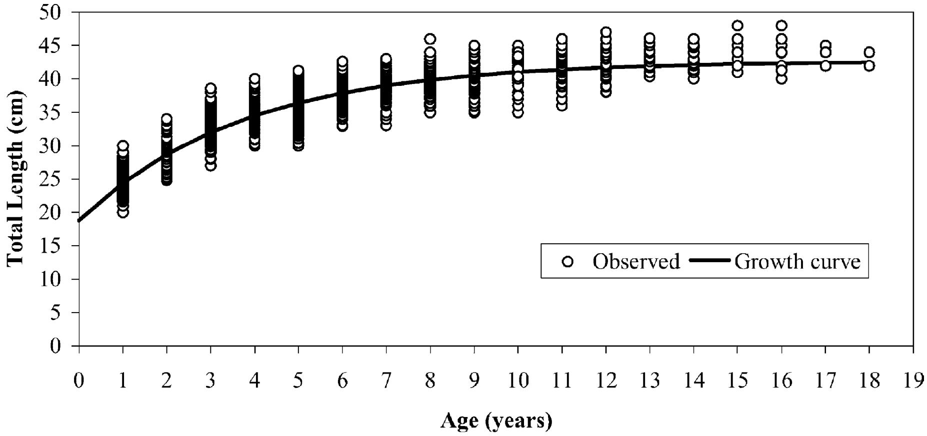
4. Discussion

Despite numerous references to the growth of northeast Atlantic mackerel, few point to a differential growth between males and females. Neja (1990) describes differences in the growth rate between sexes of mackerel in the northwest Atlantic. Females were

Table 3
von Bertalanffy growth parameters of mackerel (L_∞, k, and t_0) with standard error (S.E.), for sexes combined from data of annual mean lengths at age and pooled data 1990–2000, and for each sex from pooled data (R^2: coefficient of determination, n: number of samples)

	L_∞	S.E.	k	S.E.	t_0	S.E.	R^2	n
1990	43.6	0.429	0.241	0.013	−2.50	0.168	0.88	581
1991	44.3	0.932	0.179	0.019	−3.97	0.486	0.84	228
1992	45.7	1.783	0.122	0.024	−7.35	1.343	0.67	388
1993	42.7	0.316	0.254	0.012	−2.52	0.175	0.85	480
1994	42.1	0.306	0.297	0.013	−1.93	0.136	0.85	676
1995	43.0	0.377	0.249	0.015	−2.81	0.233	0.86	426
1996	42.6	0.277	0.288	0.012	−1.92	0.121	0.89	504
1997	43.7	0.196	0.250	0.006	−2.16	0.063	0.94	1342
1998	42.5	0.245	0.252	0.008	−2.54	0.982	0.92	1065
1999	42.1	0.254	0.295	0.010	−1.76	0.095	0.91	651
2000	42.2	0.243	0.286	0.008	−2.00	0.076	0.91	1391
1990–2000	42.7	0.089	0.268	0.003	−2.17	0.034	0.90	7732[a]
Male	42.1	0.131	0.276	0.005	−2.15	0.051	0.89	3596
Female	43.0	0.121	0.264	0.004	−2.21	0.048	0.90	3850

[a] Indetermined sex included.



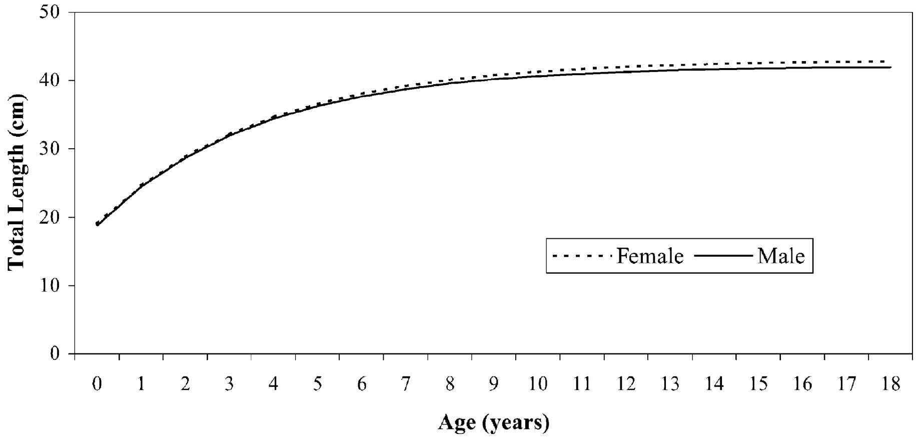


Fig. 3. von Bertalanffy growth curve fitted to observed length distributions at age data for mackerel for the 1990–2000 period (upper figure) and von Bertalanffy growth curve fitted to length distributions at age data for male and female mackerel for the 1990–2000 period (lower figure).

significantly longer than males in fish older than 7 years and the differences increased with the age of the fish. Eltink (1987) found differences in mean length at age and mean weight at age of spawning males and females of the western mackerel component in the northeast Atlantic, but attributed them to temporal differences in maturity. For local regions included in our study area, different growth between sexes has been mentioned (De la Hoz and Villegas, 1987; Lucio, 1997). Our results indicate here that in the first 3 years of life neither of the sexes dominated as to the growth rate, and the differences between lengths were not statistically significant. In fish older than 4 years, females were significantly longer than males. Significant differences were also found in the L_∞ and k of the growth curves in males and females. These differences in growth pattern may be related to the first maturity size of the mackerel. In the southern area of the northeast Atlantic, both male and female mackerel reach the first maturity after approximately 2 years, but males mature at a smaller size (Lucio, 1997). There is no evidence indicating that migration in males and females is carried out during different seasons or in different locations which would lead to their reaching different areas and affect the growth rate. Mackerel migrate in an age-length succession, but not in a male–female succession (Eltink, 1987). Plankton is the main source of food for mackerel throughout their ontogeny from larvae to adult, and as they grow, they also include fish in their diet

Table 4
Likelihood ratio test for the parameters of growth curves in mackerel males (m) and females (f)

	Base case	Coincident	L_∞	k	t_0
L_∞ (m)	42.108	42.108	42.108	42.108	42.108
k (m)	0.276	0.276	0.276	0.276	0.276
t_0 (m)	−2.147	−2.147	−2.147	−2.147	−2.147
L_∞ (f)	43.039	42.108	42.108	42.726	42.920
k (f)	0.264	0.276	0.296	0.276	0.269
t_0 (f)	−2.209	−2.147	−1.927	−2.079	−2.147
RSS_ω	22446.664	23105.024	22637.402	22470.157	22451.331
χ^2	–	215.250	63.004	7.789	1.548
d.f.	–	3	1	1	1
P	–	0.0000	0.0000	0.0053	0.2135

The base case is where two separate curves are fitted independently; the coincident column is where the lines are assumed to identical, and the remaining three columns are where the listed parameter is assumed to be equal between the two lines. RSS_ω refers to the total residual sum of squares for both curves together given constraint ω. In the base case column, 22446.664 refers to the unconstrained RSS_Ω, against which all the RSS_ω are compared. Number in all these cases is 7446 (3596 males and 3850 females).

Table 5
Comparisons of length distributions by age group of mackerel, 1990–2000

	Years	Test	1	2	3	4	5	6	7	8	9	10	11	12	13	14
	(1990–2000)	K–W	***	***	***	***	***	***	***	**	**	**	n.s.	n.s.	n.s.	n.s.
$P < 0.0051$	1990–1991	K–S	n.s.	n.s.	*	n.s.		n.s.	n.s.	n.s.	n.s.	n.s.				
	1990–1992	K–S	–	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.				
	1990–1993	K–S	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.				
	1990–1994	K–S	n.s.	n.s.	*	*	n.s.	n.s.	n.s.	n.s.						
	1990–1995	K–S	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.				
	1990–1996	K–S	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.				
	1990–1997	K–S	*	n.s.	n.s.	*	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.				
	1990–1998	K–S	n.s.	n.s.	*	n.s.	n.s.	n.s.	*	n.s.	n.s.	n.s.				
	1990–1999	K–S	*	n.s.	*	*	n.s.	*	n.s.	*	*	n.s.				
	1990–2000	K–S	–	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.				
$P < 0.0057$	1991–1992	K–S	–	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.				
	1991–1993	K–S	n.s.	n.s.	*	*	*	n.s.	n.s.	n.s.	n.s.	n.s.				
	1991–1994	K–S	n.s.	n.s.	n.s.	*	*	n.s.	n.s.	n.s.	n.s.	n.s.				
	1991–1995	K–S	n.s.	n.s.	*	n.s.	*	n.s.	n.s.	n.s.	n.s.	n.s.				
	1991–1996	K–S	n.s.	n.s.	*	n.s.	*	n.s.	n.s.	n.s.	n.s.	n.s.				
	1991–1997	K–S	n.s.	*	*	n.s.	*	n.s.	n.s.	n.s.	n.s.	n.s.				
	1991–1998	K–S	n.s.	n.s.	n.s.	n.s.	*	n.s.	n.s.	n.s.	n.s.	n.s.				
	1991–1999	K–S	*	n.s.	n.s.	n.s.	*	n.s.	n.s.	n.s.	n.s.	n.s.				
	1991–2000	K–S	n.s.	n.s.	n.s.	*	*	n.s.	n.s.	n.s.	n.s.	n.s.				
$P < 0.0064$	1992–1993	K–S	–	n.s.	n.s.	*	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.				
	1992–1994	K–S	–	*	n.s.	*	*	n.s.	n.s.	n.s.	n.s.	n.s.				
	1992–1995	K–S	–	n.s.	n.s.	*	n.s.	*	*	n.s.	n.s.	n.s.				
	1992–1996	K–S	–	n.s.	n.s.	n.s.	n.s.	*	*	n.s.	n.s.	n.s.				
	1992–1997	K–S	–	*	n.s.	n.s.	*	n.s.	*	n.s.	n.s.	n.s.				
	1992–1998	K–S	–	*	n.s.	n.s.	*	n.s.	n.s.	n.s.	n.s.	n.s.				
	1992–1999	K–S	–	*	n.s.	n.s.	*	n.s.	n.s.	n.s.	n.s.	n.s.				
	1992–2000	K–S	–		*	*	*	*	*	n.s.	n.s.	n.s.				

Table 5 (*Continued*)

	Years	Test	1	2	3	4	5	6	7	8	9	10	11	12	13	14
$P < 0.0073$	1993–1994	K–S	n.s.	n.s.	n.s.	*	n.s.	*	n.s.	n.s.	n.s.	n.s.				
	1993–1995	K–S	*	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.				
	1993–1996	K–S	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.				
	1993–1997	K–S	n.s.	n.s.	n.s.	*	n.s.	n.s.	n.s.	n.s.	n.s.	*				
	1993–1998	K–S	*	n.s.	*	*	n.s.	n.s.	*	n.s.	n.s.	n.s.				
	1993–1999	K–S	*	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.				
	1993–2000	K–S	n.s.	n.s.	*	n.s.	n.s.	*	n.s.	n.s.	n.s.	n.s.				
$P < 0.0085$	1994–1995	K–S	n.s.	*	*	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.				
	1994–1996	K–S	n.s.	*	*	*	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.				
	1994–1997	K–S	*	n.s.	n.s.	*			n.s.	n.s.	n.s.	n.s.				
	1994–1998	K–S	n.s.	*	n.s.	*	n.s.		n.s.	n.s.	n.s.	n.s.				
	1994–1999	K–S	*	*	n.s.	*	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.				
	1994–2000	K–S	*	n.s.	n.s.	n.s.		n.s.	n.s.	n.s.	n.s.	n.s.				
$P < 0.0102$	1995–1996	K–S	*	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.				
	1995–1997	K–S	*	*	*	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.				
	1995–1998	K–S	n.s.	*	*	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.				
	1995–1999	K–S	*	*	*	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.				
	1995–2000	K–S	*	*	*	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.				
$P < 0.0127$	1996–1997	K–S	n.s.	*	n	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.				
	1996–1998	K–S	*	n.s.	*	n.s.	n.s.	*	*	n.s.	n.s.	n.s.				
	1996–1999	K–S	*	n.s.	N	n.s.	n.s.	n.s.	*	n.s.	n.s.	n.s.				
	1996–2000	K–S	n.s.	*	*	n.s.	n.s.	n.s.	*	*	*	n.s.				
$P < 0.0170$	1997–1998	K–S	*	*	*	n.s.	n.s.	n.s.	*	n.s.	n.s.	n.s.				
	1997–1999	K–S	*	*	n.s.	n.s.	n.s.	n.s.	*	n.s.	n.s.	n.s.				
	1997–2000	K–S	*	n.s.	n.s.	*	n.s.	*	n.s.	n.s.	*	n.s.				
$P < 0.0253$	1998–1999	K–S	*	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.				
	1998–2000	K–S	*	*	n.s.		n.s.			n.s.	n.s.	n.s.				
$P < 0.0500$	1999–2000	K–S	*	*	n.s.	*	*	*	*	n.s.	*	n.s.				
	Sexes (1990–2000)	K–S	n.s.	n.s.	n.s.	*	*	*	*	n.s.	*	n.s.	*	n.s.	n.s.	n.s.

K–W: Kruskal–Wallis test. K–S: Kolmogorov–Smirnov test, with adjusted values for type-I error (Dunn–Sidak method for an experiment wise error rate of $\alpha = 0.05$); n.s.: not significant.

* $P < 0.05$.

** $P < 0.01$.

*** $P < 0.001$.

(e.g. Warzocha, 1988; Conway et al., 1999; Cabral and Murta, 2002). Nevertheless, neither differences in feeding behaviour between mackerel males and females nor in the gear selectivity have been reported, which prevent a sex-specific fishing targets. However, the ecological implications of the differences in growth between sexes are still unknown and could be relevant not only for the population dynamic but for the stock assessment at potential extreme levels of stock biomass.

The mackerel growth curves in the southern area (Divisions VIIIc and IXa) estimated by Gordo and Martins (1984), Cort et al. (1986), Lucio (1997) and Martins (1998) show some similarity in terms of graphics (Fig. 4). However, the related growth parameters had significant differences. Greater differences were found with the growth curves pertaining to Division IXa than with those from Division VIIIc. This may be due to the possibility that the population structure is different in each area. Division IXa presents a concentration made up chiefly of juveniles, mainly in the waters off northern Portugal (ICES, 2001), and Division VIIIc is marked by the distribution of adults (Villamor et al., 1997). Data reported by Gordo and Martins (1984) and Martins (1998) refer exclusively to Division IXa, whereas Cort et al. (1986) and Lucio

Table 6
von Bertalanffy growth parameters and observed mean length at age of the mackerel reported by different authors in northeastern Atlantic areas (ICES Divisions)

ICES area	Period of study (months)	Age (years)																L_∞	k	t_0	References
		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+				
Division VIa	1973–1976 (IV–VI)		26.0	29.9	30.8	31.9	32.3	33.1	34.0	34.3	36.4	36.2						37.27	0.221	−4.767	Kästner (1977)
Division VIa	1973–1976 (VII–VIII)			32.9	34.3	35.3	35.3	36.8	38.2	37.8	39.4	39.1						39.96	0.272	−4.293	Kästner (1977)
Divisions VIa, VIIb, VIIj)	1981 (III–VII)		27.2	28.1	32.3	33.7	34.3	36.0	37.6	37.3	37.6	37.7						39.90	0.259	−2.840	Eltkin and Gerritsen (1982)
North Sea and Western Area	1960–1985		–	–	–	–	–	–	–	–	–	–	–	–	–	–	–		0.750	−0.600	Skagen (1989)
Division IXa (Portugal coast)	1981	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	43.94	0.272	−2.404	Gordo et al. (1982)
Division IXa (Portugal coast)	1981–1983	23.0	27.5	31.6	35.2	38.1	39.5	40.6	41.3	41.9	42.2	43.0	43.3	43.5	43.7	43.9	44.0	44.36	0.268	−2.874	Gordo and Martins (1984)
Division VIIIc (Cantabrian Sea)	1982–1983 (II–VI)		24.5	29.3	32.3	33.8	35.1	38.1	39.1	40.0	40.5	41.7	41.3	41.5	41.9	42.5	43.2	43.07	0.258	−2.095	Cort et al. (1986)
Division VIIIc (Asturian coast)	1985–1986		–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	44.50	0.199	−3.180	De la Hoz and Villegas (1987)
Division VIIIabc (Basque Country)	1987–1993 (II–V)		24.7	28.7	32.4	34.7	36.7	37.9	39.7	40.7	40.9	42.1	42.1	42.6	44.1	44.7	45.6	45.88	0.196	–3.026	Lucio (1997)
Division IXa (Portugal coast)	1990–1996	23.2	27.3	31.7	34.4	35.7	37.0	37.7	39.1	40.4	41.1	42.5	42.3	42.6	44.1	46.2	44.6	44.30	0.227	−2.900	Martins (1998)
Divisions VIIIc–IXa north (Spanish coast)	1990–2000 (I–VI)		24.6	29.5	32.7	35.1	36.7	38.0	39.2	40.0	40.8	41.5	42.0	42.7	43.3	43.3	44.2	42.67	0.268	−2.173	Present work

The period of study is shown.

(1997) report on Division VIIIc. Our samples include specimens caught in IXa, although most of them come from Division VIIIc. The striking differences found as compared to Gordo and Martins (1984) could be due to the different sampling methodology used (in terms of the selectivity of the fishing gear and sampling period) or to the age interpretation applied by these authors (ICES, 1995). Our growth curves were most similar to those of Cort et al. (1986) coincident to sampling season and gears. The greatest lengths at age were estimated by Gordo and Martins (1984), that used samples taken throughout the year.

Comparing the growth curve calculated in this study with those cited for the western and northern components (Fig. 5), noteworthy differences can be seen in the graphs in comparison with the curve estimated by Kästner (1977) in waters off the west of Scotland (Division VIa). Statistically, however, the smallest differences can be seen with the growth parameters of this curve. The likelihood ratio test indicated that there was no difference between the k and L_∞ values of the two curves, implying that they follow the same growth pattern. However, the mean sizes at age were greater in the southern area.

Kästner (1977) described two groups of mackerel in the west of Scotland based on growth rates, a fast growing group and a slow growing one. Corten and Van de Kamp (1978) also described differences in the growth pattern in the Celtic Sea, indicating that western mackerel from the British Isles did not constitute a homogeneous stock.

The fast growth group follows the pattern of growth described for mackerel in the North Sea (Nedelec, 1960; Aker, 1961) and the slow growth group follows the one proposed by Steven (1952), Nedelec (1960), Molloy (1964), and Lockwood and Johnson (1976) in the Celtic Sea, northwest of Ireland and the English Channel. Nevertheless, Eltink and Gerritsen (1982), Dawson (1986) and Eltink (1987) rule out the hypothesis that there are two subpopulations in the western area and postulate an alternative explanation, stating that the growth differences are due to gradual spatial and temporal changes in the length at age during migration. The largest fish of a certain age reach spawning areas earlier and also leave for feeding areas earlier than smaller ones, which would lead to successive changes in length and weight at age. Eltink and Gerritsen (1982) calculated a growth curve for mack-

Table 7
Re-estimated parameters of the von Bertalanffy equation and growth curve comparisons

Age range compared	A			Author	B		
	L_∞	K	t_0		L_∞	K	t_0
1–10	42.7	0.280	−2.12	Kästner (1977)	38.0 n.s.	0.187 n.s.	−5.57***
				Eltink and Gerritsen (1982)	39.4**	0.258 n.s.	−3.34**
1–9	42.3	0.294	−2.01	Gordo and Martins (1984)	43.4***	0.352***	−1.82 n.s.
1–13	43.7	0.246	−2.47	Martins (1998)	45.3*	0.182**	−4.34***
1–15	44.3	0.228	−2.70	Cort et al. (1986)	43.4*	0.235 n.s.	−2.63 n.s.
				(1) Lucio (1997)	45.8***	0.196*	−3.03 n.s.
2–10	43.3	0.246	−2.67	Kästner (1977)	44.7 n.s.	0.099**	−11.45***

(A) Growth parameters obtained by fitting mean lengths at age (sexes combined 1990–2000) from this study to the von Bertalanffy equation for the corresponding age range. (B) Growth parameters obtained by fitting mean lengths at age available from the references of different authors (see Table 6) to the von Bertalanffy equation. The observed data reported by Skagen (1989), Gordo et al. (1982) and De la Hoz and Villegas (1987) are not available in the references. (1) Original data, not re-estimated. Likelihood ratio test, n.s.: not significant.

* $P < 0.05$.

** $P < 0.01$.

*** $P < 0.001$.

erel of the western area in Divisions VIIb and VIIj, which was midway between the two groups described by Kästner (1977). For this reason, they considered it to be representative of mackerel growth in the western area. Skagen (1989) estimated a growth curve using a modified von Bertalanffy equation. Using data from Norwegian catches, he concluded that mackerel caught in the North Sea are generally larger than those from the western area. From age 6 onwards the greatest lengths at age were estimated in the southern area (this study) compared with the other areas (Fig. 5). This may be attributed to the larger fish are the ones that migrate greater distances, coming to the southern area to spawn. In addition to the age-length succession in the migratory routes of the mackerel, the interpretation of growth will be dependant upon the season when samples are obtained and seasonal differences in growth were reported (Castelló and Hamre, 1969). Population samples that are not representative will cause differences in growth (Dawson, 1986). In migratory species like the mackerel, input data in growth studies may often be biased (elements might be missing as the whole

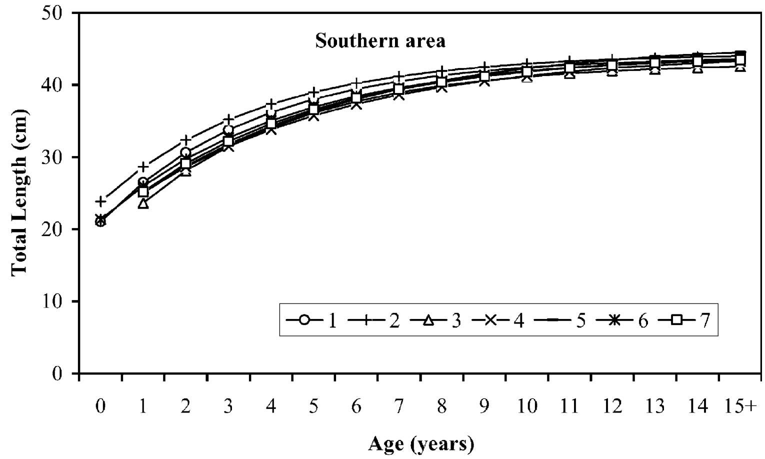


Fig. 4. Theoretical growth rate of the southern component of northeast Atlantic mackerel, according to the von Bertalanffy equation calculated by different authors: 1. Gordo et al. (1982), 2. Gordo and Martins (1984), 3. Cort et al. (1986), 4. De la Hoz and Villegas (1987), 5. Lucio (1997), 6. Martins (1998), 7. this study. See Table 6 for the respective geographical areas for each reference.

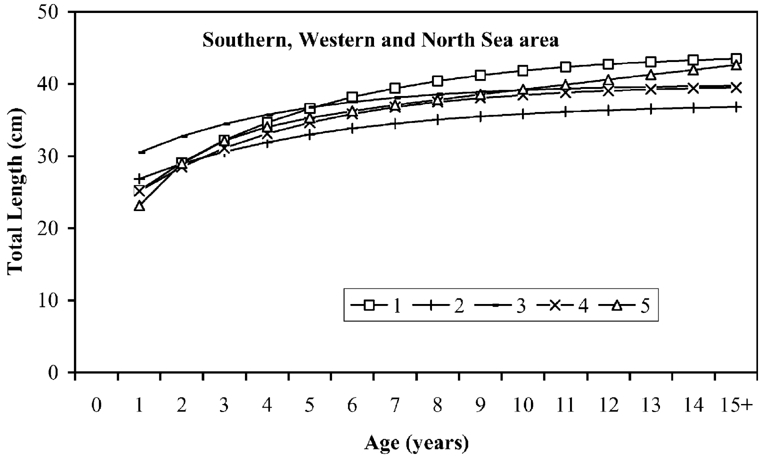


Fig. 5. Theoretical growth rate of northeast Atlantic mackerel in different areas, according to the von Bertalanffy equation calculated by different authors: 1. southern area (Division VIIIc and Subdivision IXa north) (this study), 2. western area (Division VIa) (Kästner, 1977), 3. western area (Division VIa) (Kästner, 1977), 4. western area (Divisions VIIb and VIIj) (Eltink and Gerritsen, 1982), 5. northern and western area (Skagen, 1989).

cohort may not be present in the area where the samples are taken) (Sparre and Venema, 1992).

Fish growth is characterised by its variability and the same species may have different growth patterns. During the period of study (1990–2000) interannual variability was detected between the mackerel length distributions of some age groups (see Table 5), although when viewed graphically the differences were not very noticeable. The length distribution of an age class is partly determined by the temporal difference (during the spawning season) of the moment when the larvae hatch, food availability, environmental conditions, and time of year sampled. All these factors may give rise to different growth rates (Wootton, 1990). In the area of study, primary production is, to a great extent, determined by upwelling processes (Blanton et al., 1984; Lavín et al., 1998) which may have a major effect on juvenile growth (Cushing, 1995). During their migration, adults are subject to highly variable environmental conditions, which influence their growth and spawning potential (Lam, 1983; Weatherley and Gill, 1989).

The annual growth rate of northeast Atlantic mackerel may be influenced by environmental and population factors (variation in spawning period, the composition at age, migration of specific lengths or a combination of these factors) as was reported for other fish (Parrish et al., 1985). The spawning season occurs earlier in the south of Europe than in the north (Iversen, 1981; Lockwood et al., 1981; Jorge et al., 1982; Solá et al., 1990). Referring to the spawning season, Dawson (1991) found that the radius of the first otolith ring was greater in mackerel from the south of the Bay of Biscay. The catch composition at age also varies in the different areas of the northeast Atlantic (ICES, 1996–2001). As previously stated, Dawson (1986) and Eltink (1987) linked differences in growth to gradual spatio-temporal changes in length at age during the migration of mackerel through European waters. Another cause of variability in mackerel growth rate may be the effect of population density, as found by Mackay (1973), Overholtz (1989) and Neja (1995) in the northwest Atlantic, especially in the youngest ages. Agnalt (1989) also found in North Sea mackerel a negative correlation between mean fish length at 1 and 2 years of age and stock biomass in the 1970s. High levels of fishing mortality may change the population structure and growth rates. However, northeast Atlantic mackerel has not shown great variations in year classes in the last 15 years (ICES, 2001) and the compensatory process are difficult to investigate. The spatio-temporal variability in the growth pattern of north Atlantic mackerel need to be taking into account in the stock assessment. To minimise the associated uncertainties the procedures should include: (1) age determination through the analysis of hard structures such as otoliths, because the use of length distributions for age assignation is questionable (Skagen, 1989); (2) routine ageing and growth parameters determination; (3) coupling of growth parameters with referenced area and time period; (4) integration of the growth parameters from different mackerel components or from specific local areas.

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